

Weak-Signal Architecture for Healthcare Monitoring and Early-Warning Triage

Toward a Stratified, Governance-Aware Architecture for Early Clinical Deterioration Handling

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Abstract

Healthcare monitoring and early warning triage often begin not with a single obvious event, but with weak, partial, noisy, and locally ambiguous signals: mild vital-sign drift, subtle responsiveness change, weak symptom inconsistency, small laboratory deviation, repeated departure from a recent baseline, or measurement disagreement across channels. In such cases, the architectural problem is not diagnosis, but promotion control: how weak findings should influence attention before they are allowed to become stronger triage consequence. This bridge paper proposes a minimal, implementable transfer contract for early-warning handling in clinical monitoring: (1) a light state ladder (H0–H4, R0, Q0), (2) a compact event object schema, (3) a deterministic promotion predicate framework, and (4) a promotion-record format for audit and review. The contribution is architectural rather than clinical. The paper provides concrete examples and a short checklist for governance and auditability; it does not propose diagnostic models, treatment recommendations, clinical workflows, medical-device efficacy, or replacement of clinicians.

1. Introduction

Healthcare monitoring rarely begins with perfect certainty. A patient may show mild heart-rate drift, a slight respiratory change, a small temperature irregularity, inconsistent symptom reporting, a weak laboratory deviation, a subtle responsiveness shift, or repeated departure from a recent corridor of stability. Any one of these signals may be temporary, benign, or measurement-related. Any one of them may also be an early sign that the current cheap reading is no longer fully reliable.

The architectural problem is promotion control: how to preserve and act on weak, structured deviations without prematurely creating durable triage states or producing false alerts.

This paper takes a narrow position. Healthcare monitoring and early warning triage are appropriate application domains for an architecture that separates bounded observation, provisional interpretation, short-horizon salience escalation, and governed consequence. The point is not automated diagnosis or medical authority. The point is disciplined escalation under uncertainty in systems where weak signals may be clinically important before they become certain.

The contribution is architectural rather than clinical. This paper does not prescribe a diagnostic model, treatment pathway, device implementation, regulatory claim, or benchmark. It argues something smaller and more portable: monitoring systems need a disciplined middle layer in which weak early-warning signals can raise attention before they are allowed to become stronger triage consequence.

2. Why Healthcare Monitoring Needs a Middle State

Clinical deterioration often develops as drift rather than as a single clean event. Vital signs may move gradually. Responsiveness may change subtly. A symptom pattern may become less consistent with the recent corridor. A laboratory value may be weakly abnormal but not decisive. A monitor may produce an unusual reading that is poorly corroborated. A caregiver may report that the patient seems slightly different without being able to name a strong diagnostic category.

At that stage, the system does not yet possess a diagnostic fact. What it possesses is a local precursor pattern. That pattern may be meaningful, but it remains incomplete. The architecture therefore needs a disciplined middle state between raw observation and stronger triage consequence.

Without such a middle state, monitoring systems tend to collapse in one of two directions. In the first, weak anomalies are treated as ordinary noise until they become obvious, losing lead time and context. In the second, weak irregularities are pushed too quickly into alerts or stronger triage states, increasing noise, burden, and distrust. A better architecture preserves provisional significance without premature commitment.

A disciplined middle state preserves provisional significance for weak precursor patterns without committing to diagnosis or durable triage. This middle state is not a substitute for clinical judgment. It is a boundary discipline that allows weak signals to matter at the level of attention, review, and evidence gathering without allowing them to silently become diagnosis, treatment instruction, or durable clinical state.

3. Core Claim and Design Principle

The central claim of this paper is simple:

Healthcare monitoring and early warning triage are suitable application domains for a weak-signal architecture in which bounded findings raise attention first, candidate meanings remain provisional, and stronger consequence occurs only through persistence, convergence, and governance.

This architecture complements existing monitoring and clinical workflows by inserting an explicit, inspectable layer for provisional attention. In many monitoring settings, the first useful evidence is too partial to deserve strong commitment but too structured to deserve indifference.

The governing design principle is equally simple:

Weak clinical or monitoring anomalies should influence attention before they justify stronger triage consequence.

That means a small vital-sign drift, weak symptom inconsistency, subtle behavioral change, or measurement mismatch can justify tighter observation, repeated measurement, shorter review horizon, or clinician attention flag without immediately becoming a diagnosis, treatment recommendation, or durable escalation state.

This is not a claim that weak signals are secretly decisive. It is a claim that they are often useful earlier than hard-threshold logic allows. A disciplined architecture should preserve that usefulness while still demanding stronger support for stronger consequence.

4. Stratified Architecture Mapping

The architecture has four processing layers: (1) Bounded Observation — emit narrow, local findings; (2) Provisional Interpretation — attach candidate meanings while preserving ambiguity; (3) Short-Horizon Salience Escalation — tighten observation and request corroboration without authorizing durable state; (4) Governed Consequence — explicit, predicate-bound promotion into triage workflows.

Lifecycle: notice → interpret → shape → govern.

4.1 Bounded Observation

Emits source-scoped findings such as mild heart-rate irregularity, slight respiratory-rate change, modest oxygenation fluctuation, small temperature shift, subtle responsiveness change, weak symptom inconsistency, small laboratory deviation, repeated departure from a recent baseline, or disagreement across measurement sources. This layer does not claim deterioration, infection, respiratory compromise, dehydration, sepsis, or any other clinical condition.

4.2 Provisional Interpretation

Forms candidate local meanings such as possible early deterioration, possible transient instability, possible measurement artifact, possible hydration-related concern, possible respiratory concern, possible infection-related concern, possible medication or timing effect, possible recovery fluctuation, or unresolved anomaly needing more evidence. Candidate meanings are admissible hypotheses, not final truths.

4.3 Short-Horizon Salience Escalation

Permits reversible attention shifts: tighter monitoring, shorter observation windows, repeated measurement, broader review of nearby indicators, temporary watch state, or clinician attention flag. This layer remains temporary and non-authorizing. It does not write durable clinical state,

mutate long-horizon baselines, trigger treatment recommendation, or authorize stronger triage consequence.

4.4 Governed Consequence

Permits stronger consequence only after declared support builds. Possible outputs include elevated triage watch level, clinician review routing, additional assessment recommendation, stronger monitoring status, or bounded escalation pathway. Promotion must be explicit, inspectable, and predicate-bound rather than silently triggered by local salience alone.

5. Minimal Operational Contract

This section keeps the transfer concrete without turning the paper into a medical-device specification or clinical workflow design. The point is not to prescribe one implementation. It is to define a minimal contract surface that makes the architecture inspectable and portable.

5.1 Light State Ladder

A minimal state ladder is enough to distinguish weak-signal attention from stronger triage consequence:

- **H0 — Expected Corridor:** observations remain within the expected corridor for the current patient context, monitoring regime, or recent baseline.
- **H1 — Noticed:** a weak local deviation is recorded as a bounded finding.
- **H2 — Watching:** the deviation persists, recurs, or appears near other weak departures within a bounded short horizon.
- **H3 — Corroborated:** two or more independent indicators converge, or a credible corroborator strengthens the concern.
- **H4 — Escalated:** a governed triage-review or stronger monitoring boundary has been crossed.
- **R0 — Resolved:** the deviation returns to corridor without stronger support.
- **Q0 — Quarantined:** evidence favors measurement artifact, instrumentation issue, documentation mismatch, or another non-clinical explanation.

The key boundary is simple: H1–H3 are attention and evidence states; H4 belongs to governed triage workflow.

Operational semantics below are example defaults, not clinical thresholds:

- H1 — Noticed: single bounded finding recorded; no corroboration; retention window = 30 minutes.
- H2 — Watching: same finding persists or a second weak finding appears within 30 minutes.

- H3 — Corroborated: two or more independent channels corroborate within 30 minutes, or one channel shows confidence ≥ 0.80 across three consecutive samples.
- H4 — Escalated: promotion predicates are satisfied and a promotion record is emitted for governance review.

5.2 Minimal Event Object

A weak-signal architecture needs a small event object that can preserve structure without forcing immediate alert semantics. In minimal form, a healthcare monitoring event should carry:

- event identity and time bounds
- patient or case scope identifier
- source channel or observation type
- recent baseline or expected corridor reference
- observed value, change, or qualitative finding
- deviation method or comparison basis
- current state in the ladder
- supporting evidence or corroborators
- short-horizon actions taken
- governance or routing notes where relevant
- closure reason if resolved or quarantined

The point is not to freeze a universal schema. The point is to preserve enough context that weak deviations remain reviewable, comparable, and governable.

A compact typed event object may look like this:

```
{
  "event_id": "string",
  "patient_id": "string",
  "time_window": "ISO8601 interval",
  "source_channels": ["string"],
  "baseline_reference": "string",
  "finding_summary": "string",
  "observed_values": [
    {
      "channel": "string",
      "value": 0.0,
      "unit": "string",
      "timestamp": "ISO8601",
      "confidence": 0.0
    }
  ]
}
```

```

    }
  ],
  "candidate_interpretations": ["string"],
  "ladder_state": "H0|H1|H2|H3|H4|R0|Q0",
  "supporting_evidence": ["evidence_id"],
  "short_horizon_action": "string",
  "promotion_status": "not_promoted|promoted",
  "closure_reason": "string|null"
}

```

Fields are typed. Confidence is a 0–1 source-channel score. Each `evidence_id` links to raw measurement records, notes, or derived observations for audit.

5.3 Transition Logic

The transition logic should remain explicit:

- $H0 \rightarrow H1$ when a bounded finding departs from the local corridor enough to be recorded.
- $H1 \rightarrow H2$ when the finding persists, recurs, or appears near another weak departure within a declared short horizon.
- $H2 \rightarrow H3$ when independent corroboration strengthens the candidate concern.
- $H3 \rightarrow H4$ only when declared promotion predicates are satisfied and a rationale-bearing transition record is emitted.
- $H1/H2/H3 \rightarrow R0$ when the pattern dissipates without stronger support.
- $H1/H2/H3 \rightarrow Q0$ when evidence favors measurement artifact, instrumentation issue, documentation mismatch, or another non-clinical explanation.

Promotion function (deterministic example):

Let $P = [\text{persistence, recurrence, convergence, proximity, baseline_departure, corroboration_quality}]$. Each predicate is boolean. Promote to H4 if $\text{sum}(P_weighted) \geq \text{threshold}$, where weights and threshold are governance-configurable. Example default:

```

weights = {
  persistence: 2,
  recurrence: 1,
  convergence: 3,
  proximity: 1,
  baseline_departure: 1,
  corroboration_quality: 2

```

```

threshold = 5

```

Under this example, persistence + convergence is sufficient for promotion only when confidence and corroboration quality are adequate under local governance rules. Persistence alone is insufficient. Saliency alone is insufficient. Every promotion emits a promotion record containing the predicate vector, timestamps, supporting evidence, and reviewer boundary.

6. Promotion Predicates and Governance

Promotion control is the core of the healthcare transfer. A weak signal may be important, but importance is not the same as authority. Promotion therefore requires an explicit predicate vector and an audit-bearing transition record.

6.1 Predicate List

The following predicates are example defaults, not clinical thresholds. Their purpose is to show how promotion can be made deterministic and inspectable.

Persistence

Definition: The same deviation remains present across the review window.

Required inputs: observed_values, timestamps, baseline_reference.

Boolean rule: true if $\text{count}(\text{samples with deviation}) \geq 2$ within window W.

Recurrence

Definition: The deviation returns after temporary resolution or reappears under comparable conditions.

Required inputs: observed_values, prior event state, timestamps.

Boolean rule: true if the event re-enters H1 or H2 after R0 within recurrence window W2.

Convergence

Definition: Independent channels support the same candidate concern.

Required inputs: source_channels, candidate_interpretations, supporting_evidence.

Boolean rule: true if two or more independent channels support the same candidate within W.

Proximity

Definition: A weak finding appears near another relevant change in time, channel, or patient context.

Required inputs: timestamps, channel scope, case context.

Boolean rule: true if a second relevant weak finding occurs within proximity window X.

Baseline departure

Definition: The finding departs from the recent local corridor.

Required inputs: observed_values, baseline_reference, deviation method.

Boolean rule: true if deviation exceeds the locally declared corridor boundary.

Corroboration quality

Definition: Support is strong enough to reduce single-channel artifact risk.

Required inputs: confidence scores, source independence metadata.

Boolean rule: true if corroborating evidence has confidence greater than or equal to the configured minimum and is not a same-source duplicate.

6.2 Promotion Record

A minimal promotion record should use exact, reviewable fields:

```
{
  "promotion_id": "string",
  "event_id": "string",
  "prior_state": "H1|H2|H3",
  "new_state": "H4",
  "predicate_vector": {
    "persistence": true,
    "recurrence": false,
    "convergence": true,
    "proximity": false,
    "baseline_departure": true,
    "corroboration_quality": true
  },
  "weighted_score": 0,
  "threshold": 0,
  "supporting_evidence_ids": ["evidence_id"],
  "evaluation_window": "ISO8601 interval",
  "promoting_actor": "system|human",
  "review_boundary": "clinician_queue|nurse_review|local_policy_review|none",
  "rationale_text": "string",
  "uncertainty_notes": "string",
  "audit_id": "string",
  "reversal_policy": "string"
}
```


6.3 Governance Checklist

- Promotion must produce a promotion record with `prior_state`, `new_state`, `predicate_vector`, `supporting_evidence_ids`, `timestamps`, `promoting_actor`, `rationale_text`, and `audit_id`.
- Promotions triggered automatically must be logged and routed to the configured human-review boundary when local policy requires review.
- Promotion does not imply diagnosis, treatment recommendation, or final clinical state. It means a governed review or stronger monitoring boundary has been crossed.
- Reversal policy must be declared. Promotions should decay to R0 if corroboration fails or the pattern returns to corridor within a configured decay window.
- Q0 quarantine must remain available when evidence favors measurement artifact, instrumentation issue, documentation mismatch, or another non-clinical explanation.

This keeps escalation reviewable. It also protects against two opposite failures: alarm collapse, where every weak signal becomes an alert, and indifference collapse, where weak precursor structure disappears because it is not yet definitive.

7. Worked Examples

7.1 Mild Vital-Sign Drift Timeline

This example uses synthetic values only. It illustrates event lifecycle and predicate evaluation, not clinical thresholds.

09:00 — H0: HR=78 bpm, RR=16/min, SpO2=98%. Values remain inside the recent local corridor.

09:05 — HR=92 bpm, confidence=0.70. The value departs the local corridor. Create event E1 with `ladder_state=H1_Noticed`.

09:10 — HR=94 bpm and RR=18/min. Update E1 to H2_Watching. Predicate `persistence=true` because the heart-rate deviation appears in two samples within the 30-minute review window. Predicate `convergence=false` because the respiratory shift remains weak and not yet independently sufficient.

09:15 — SpO2 decreases by 1 percentage point from recent corridor and respiratory shift persists. Update E1 to H3_Corroborated. Predicate `convergence=true` because a second channel now supports the same candidate concern. Predicate `corroboration_quality=true` if source confidence and independence rules are met.

09:16 — Promotion evaluation: `predicate_vector = {persistence:true, recurrence:false, convergence:true, proximity:false, baseline_departure:true, corroboration_quality:true}`.

Weighted score = $2 + 3 + 1 + 2 = 8$. Threshold = 5. Emit promotion record and route to configured clinician-review queue. State becomes H4_Escalated.

09:30 — Review or observation resolves concern as transient under local workflow. closure_reason="resolved after observation". State moves to R0.

7.2 Measurement Ambiguity

A monitoring system receives a weak abnormal reading with inconsistent corroboration. The pattern may reflect early instability, but it may also reflect instrumentation artifact, patient movement, poor signal quality, documentation timing, or local measurement noise. The architecture preserves the anomaly, requests or awaits repeat measurement, compares nearby indicators, and holds the case in a provisional state. If support strengthens, promotion may occur. If evidence favors artifact, the case moves to Q0 rather than becoming either a false alert or invisible noise.

7.3 Subtle Responsiveness Change

A patient shows a small responsiveness shift, mild symptom inconsistency, and repeated departure from a recent behavioral corridor. The architecture preserves candidate meanings provisionally, raises attention, and keeps ambiguity open. The system may shorten the review window, invite clinician attention, or request additional assessment, but it does not treat the weak pattern as diagnosis or treatment instruction. Escalation depends on persistence, convergence, and governed review.

8. Relationship to the Broader Weak-Signal Stack

The stack reuses standard weak-signal primitives adapted to clinical monitoring constraints: bounded finding, provisional candidate, short-horizon salience, and governed promotion.

Reflex-style attention explains why short-horizon salience can rise without memory writes, baseline updates, or stronger consequence. Constraint-deviation logic explains why deviations should be compared against scoped baselines and routed through event formation rather than collapsed into one opaque alarm. Experience-entry and memory-boundary logic explain why represented events should not automatically become durable commitments. Governance logic explains why stronger consequence requires explicit promotion criteria, rationale, and review boundaries.

The resulting stack is simple:

bounded finding → provisional candidate → short-horizon attention → governed promotion

This is the same practical logic as:

notice → interpret → shape → govern

9. Scope and Non-Goals

This paper is intentionally limited.

It is not a diagnostic paper.

It is not a treatment recommendation paper.

It is not a medical-device specification.

It is not a clinical-efficacy claim.

It is not a regulatory submission.

It is not a benchmark report.

It is not a replacement for clinicians, nurses, caregivers, or existing clinical workflows.

It does not define disease-specific thresholds, patient-risk models, treatment pathways, medication decisions, or emergency-response policy.

The claim is architectural: healthcare monitoring systems need a disciplined place for weak early-warning signals to matter before they harden into alerts, triage states, or ignored noise.

10. Practical Value

Practical value: this architecture is intended to reduce both false escalation and missed early deterioration by separating attention from authority and making promotions auditable and reversible. A weak finding may be recorded without becoming a diagnosis. A candidate interpretation may remain active without becoming durable state. Short-horizon salience may rise without authorizing treatment or strong triage. Stronger consequence may occur only when declared support has accumulated. Empirical validation would still be required before any clinical or operational claim.

11. Conclusion

This paper proposes a minimal, implementable transfer contract for weak-signal handling in clinical monitoring: a light state ladder, a typed event object, deterministic promotion predicates, a promotion-record format, and a governance checklist. The bridge is narrow: weak clinical or monitoring anomalies should influence attention before they justify stronger triage consequence.

Weak signals do not need to be ignored or overread. They can be noticed locally, interpreted provisionally, shaped into bounded attention, and governed before persistence or consequence.